Chapter 7 — Emerging fairness topics

Fairness challenges evolve as models become generative, training becomes decentralized

Learning objectives

- Connect future methods with the lifecycle practices developed throughout the chapter

Bias in generative AI

- Uneven representation
- A text-to-image model may repeatedly depict leaders or doctors from one demographic group
- Open-ended outputs make fairness evaluation broader than checking one fixed classification

Evaluating and mitigating generative bias

- Evaluation should use diverse prompts, repeated samples, intersectional groups
- Deployment safeguards
- Because interventions can shift or conceal harms

Fairness in federated learning

- Federated learning trains across decentralized devices or organizations without
- Privacy benefits do not guarantee fairness
- Participants may contribute very different amounts and distributions of data

Fairness-aware federation

- Reporting performance for each participating population
- Teams must also examine which clients are repeatedly excluded and whether privacy
- The objective should balance global utility with acceptable performance for less

Regulation and deployment

- Privacy, anti-discrimination, consumer protection
- Deployment controls rather than assembled after a complaint

Future directions

- Bias-aware datasets, causal methods, fairness-aware optimization, explainability
- Their value depends on clear problem definitions and real-world validation
- Ongoing monitoring

Takeaways

- Generative systems expand fairness evaluation to open-ended content and intersectional
- Federated systems introduce uneven participation and local performance concerns
- Regulation makes documentation, testing, oversight
- Across all settings, fairness remains a lifecycle practice rather than a one-time metric

Next

- Complete the quiz for this part
- Emerging challenges